

edureka!
a Veranda Enterprise

ILLINOIS TECH

POST GRADUATE PROGRAM IN **GENERATIVE AI AND ML**

**Applied Machine Learning
Techniques**



Module Outline

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**ML: Data Preprocessing and
Feature Engineering**

**Supervised Learning -
Regression**

**Supervised Learning -
Classification**

Unsupervised Learning

**Clustering Algorithms and
Dimensionality Reduction**

Ensemble Learning

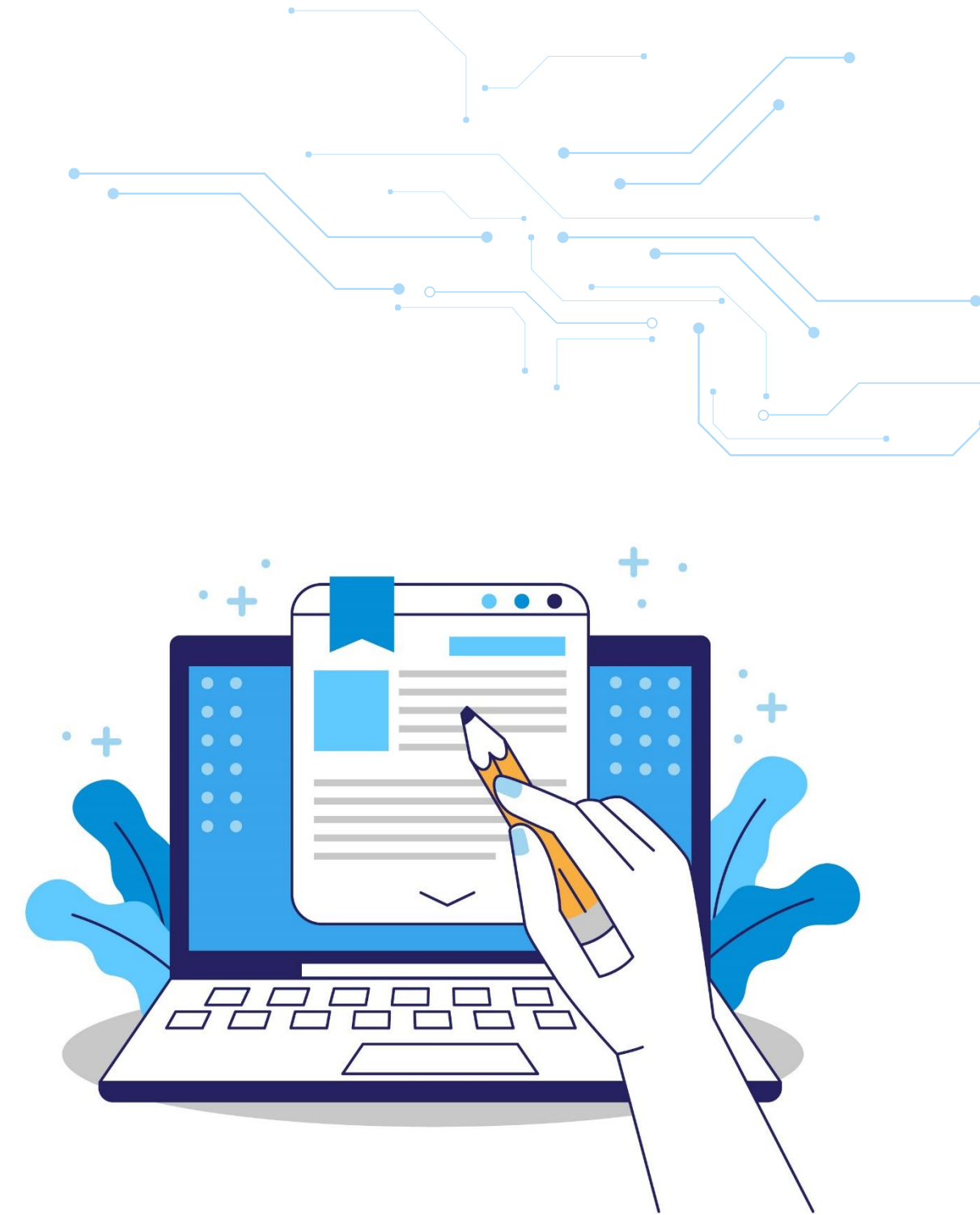
**Time Series Analysis and
Reinforcement Learning
(Self Paced)**

**Recommender Systems
(Self Paced)**

Machine Learning : Data Pre-processing and Feature Engineering

Topics

- e! What is Machine Learning?
- e! Scope and Importance of Machine Learning
- e! Types of Machine Learning
- e! Core Components of Machine Learning
- e! History and Evolution of Machine Learning
- e! Machine learning applications and key terms
- e! Machine Learning Pipeline
- e! Data Preprocessing
 - Handling Missing Data
 - Detecting and Handling Outliers
 - Encoding Categorical Values
 - Feature Scaling
 - Train Test Split
- e! Exploratory Data Analysis
 - What is EDA?
 - Mean, Median and Mode
 - Imputation
 - Univariate, Bivariate and Multivariate Analysis
 - Detecting and Handling Data Imbalance
- e! Feature Engineering
- e! Feature Selection and Methods
- e! Handling Multicollinearity
- e! Feature Transformation



Learning Objectives

By the end of this lesson, you will be able to:

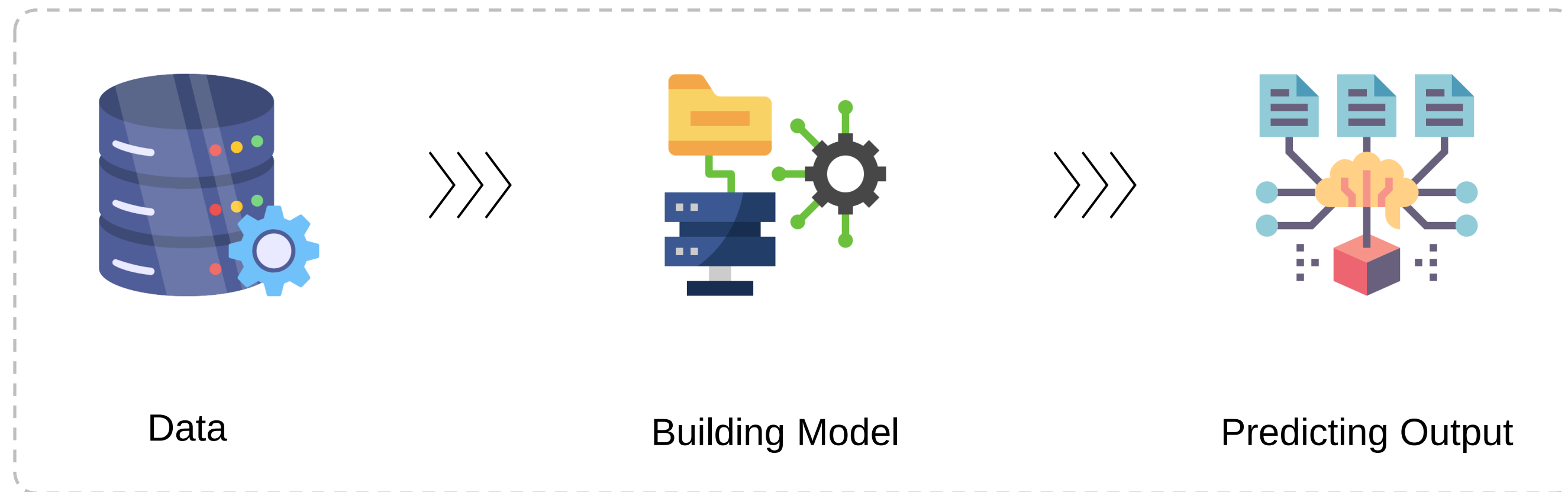
- e! Define Machine Learning, its types, and key terms.
- e! Explain the importance, scope, and real-world applications of Machine Learning.
- e! Perform data preprocessing tasks such as handling missing data, encoding categorical values, and feature scaling.
- e! Evaluate datasets using Exploratory Data Analysis (EDA) and detect issues like outliers and data imbalance.
- e! Apply feature engineering techniques such as feature selection, transformation, and handling multicollinearity.



Introduction to Machine Learning

What is Machine Learning?

Machine learning is a subset of artificial intelligence (AI) that allows systems to learn and improve automatically from experience without explicit programming.



Scope of Machine Learning



Innovation and Competitive Edge

Drives AI advancements, smart systems, and business growth.

Real-Time Processing

Enables fraud detection, autonomous systems, and instant decision-making.

Healthcare and Science

Advances medical diagnostics, drug discovery, and disease prediction.



Automation and Efficiency

Reduces manual effort, increases speed, and optimizes workflows.

Data-driven Decisions

Extracts insights, improves forecasting, and enhances business strategies.

Personalization

Powers recommendation systems, chatbots, user-specific experiences.

Importance of Machine Learning

Automation

With the help of machines, a few routine tasks are automated daily, like data processing and spam filtering.



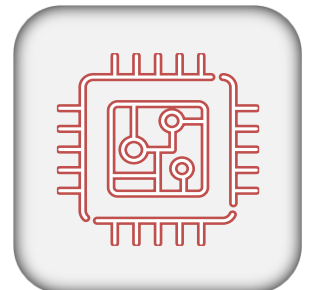
Time saving

Rather than creating an algorithm from scratch for a particular task, it is easier to train available models and save a lot of time.



Computation Power

With GPUs, it is quite easy to train machine learning models on massive amounts of data.



ML vs. Traditional Programming

Aspect	Traditional Programming	Machine Learning
Approach	Rule-based, explicit logic.	Learns patterns from data.
Development	Requires manually coding all rules.	Trains on data to make predictions.
Flexibility	Static – must update rules manually.	Adaptive – improves with more data.
Handling Complexity	Struggles with large, unstructured data.	Excels in complex, pattern-based problems.
Examples	Sorting algorithms, tax calculations, basic automation.	Image recognition, fraud detection, NLP, self-driving cars.

Types of Machine Learning



**Supervised
Machine Learning**



**Unsupervised
Machine Learning**

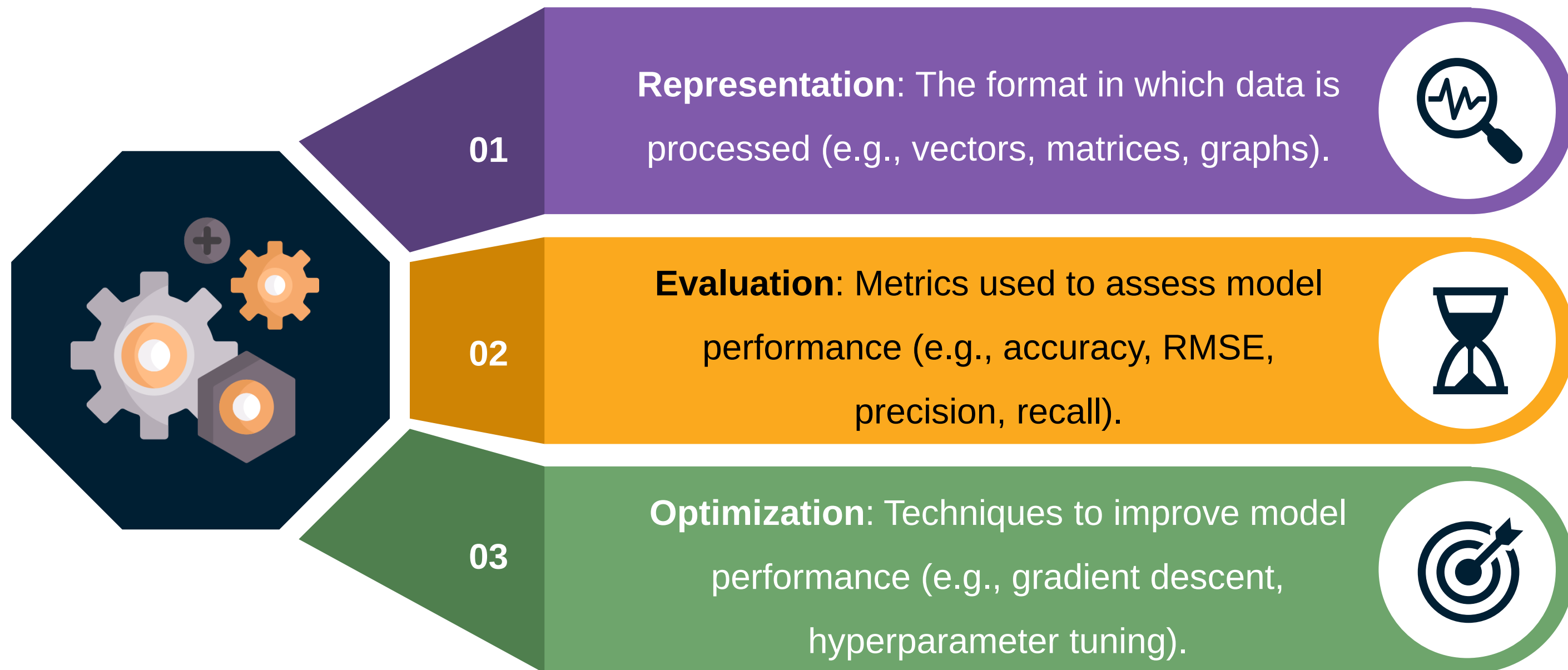


**Reinforcement
Machine Learning**

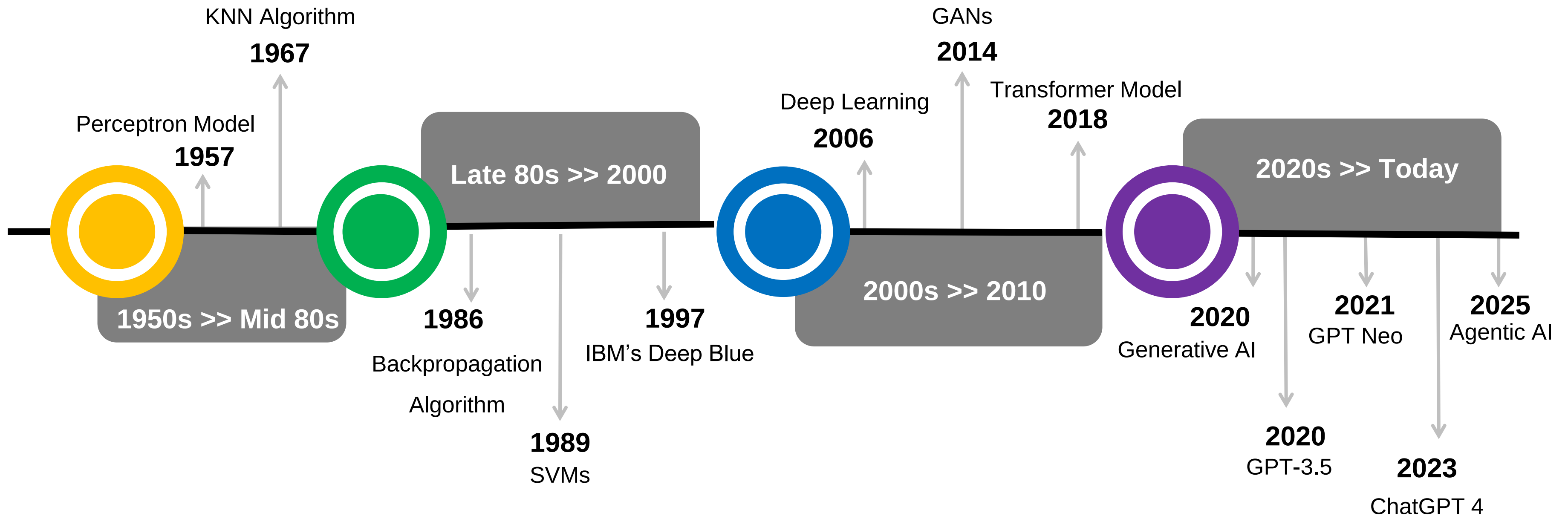


**Semi Supervised
Machine Learning**

Core Components of Machine Learning



History and Evolution of Machine Learning



Machine Learning Applications



Cybersecurity

Intrusion detection,
malware classification.

Manufacturing

Predictive maintenance,
quality control.



Healthcare

Disease diagnosis, medical
imaging, drug discovery.

Finance

Fraud detection, risk
assessment, algorithmic trading.

Retail

Customer segmentation,
recommendation systems,
demand forecasting.

Machine Learning Key Terms



01

Features

Independent variables or attributes used for prediction.



02

Labels

Target values in supervised learning.



03

Model

A mathematical representation trained on data to make predictions

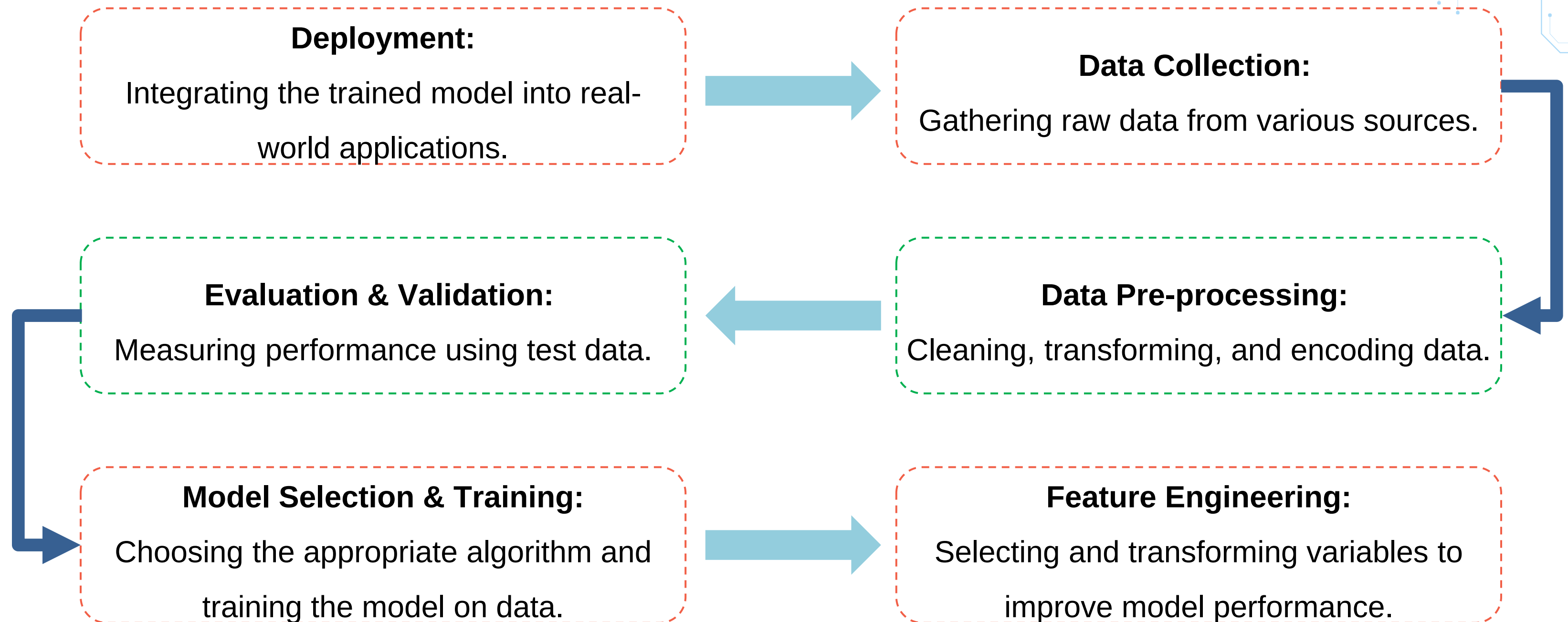


04

Training Data

The dataset used to train the ML model.

Machine Learning Pipeline



Prerequisites for Machine Learning



Mathematics and Statistics

Linear algebra, calculus, probability, and statistics.



Programming Skills

Python, R, or Java.



Data Handling

Familiarity with databases and data processing tools.

Data Preprocessing

Importance of Data Quality

Improves Model Accuracy

Clean and relevant data leads to better generalization.

Reduces Bias and Variance

Inconsistent or incomplete data can introduce bias.

Enhances Model Interpretability

Well-structured data improves decision-making.

Ensures Robustness

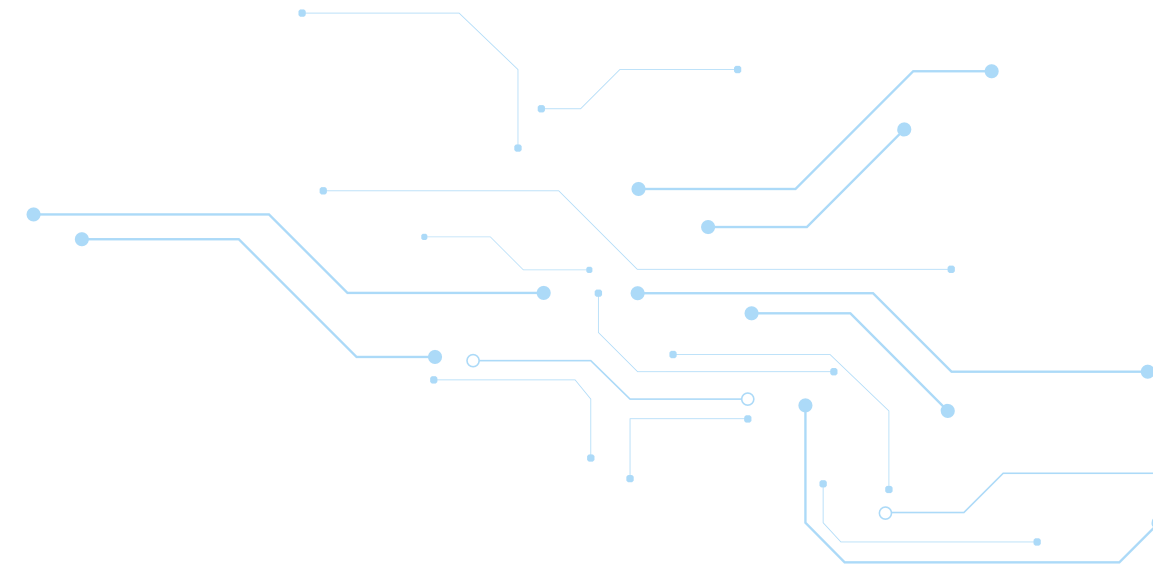
High-quality data helps models perform well on real-world scenarios.

Prevents Overfitting

Noisy data can mislead the model into learning irrelevant patterns.



Handling Missing Data



Before

	Customer_ID	Age	Income	Email	Feedback	Source
0	101.0	25.0	50000	a@example.com	Good	Internal
1	102.0	34.0	60000	b@example.com	Bad	External
2	103.0	45.0	70000	c@example.com	Excellent	Internal
3	104.0	29.0	NaN	d@example.com	??	External
4	105.0	NaN	80000	e@example.com	Average	Internal
5	106.0	40.0	90000	None	Poor	External
6	106.0	40.0	90000	f@example.com	Poor	External
7	107.0	50.0	100000	g@example.com	Good	Internal
8	108.0	60.0	110000	h@example.com	Bad	External
9	NaN	200.0	??	h@example.com		Internal

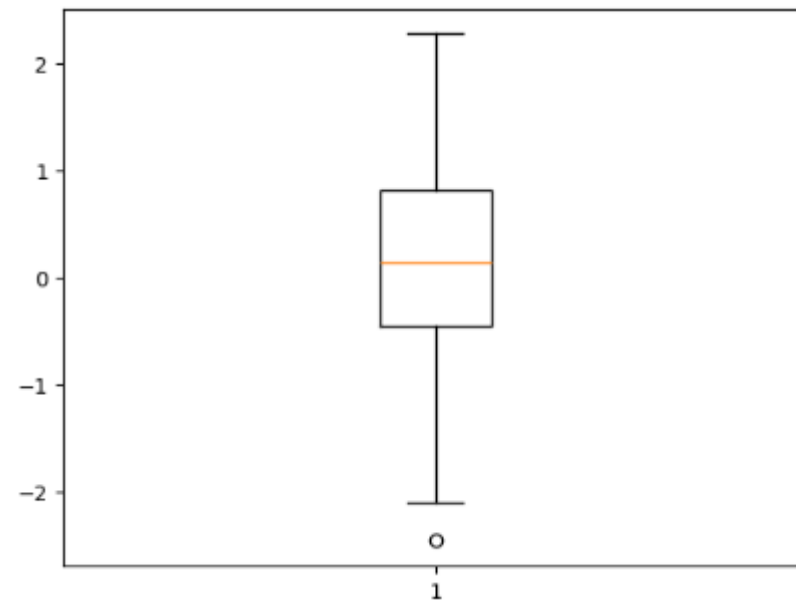
After

	Customer_ID	Age	Income	Email	Feedback	Source
0	101.0	25.0	50000.0	a@example.com	Good	Internal
1	102.0	34.0	60000.0	b@example.com	Bad	External
2	103.0	45.0	70000.0	c@example.com	Excellent	Internal
3	104.0	29.0	81250.0	d@example.com	??	External
4	105.0	40.0	80000.0	e@example.com	Average	Internal
5	106.0	40.0	90000.0	None	Poor	External
6	106.0	40.0	90000.0	f@example.com	Poor	External
7	107.0	50.0	100000.0	g@example.com	Good	Internal
8	108.0	60.0	110000.0	h@example.com	Bad	External
9	108.0	200.0	81250.0	h@example.com		Internal

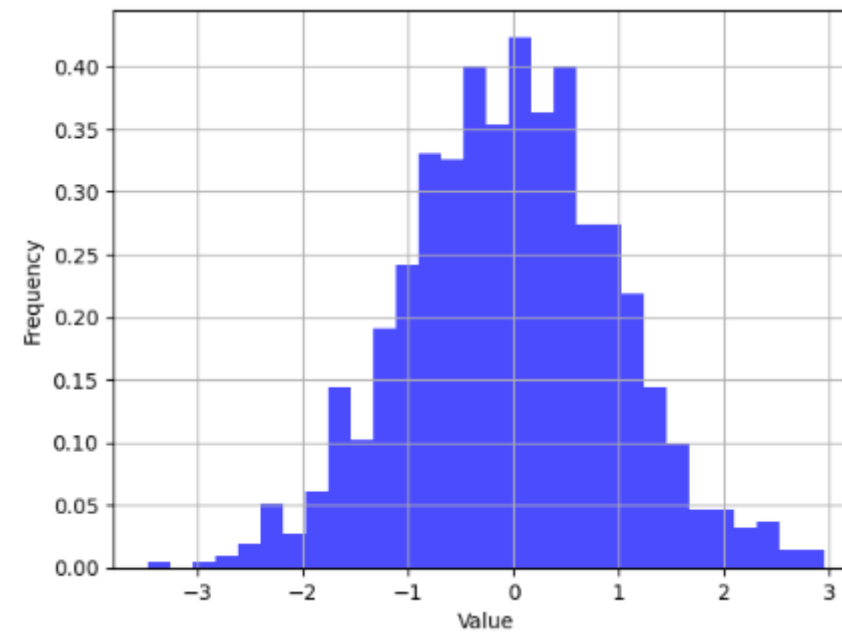
Mean/Median Imputation: Filling missing values with the mean or median of the column.

Dropping Rows/Columns: Removing missing data if the percentage of missing values is too high.

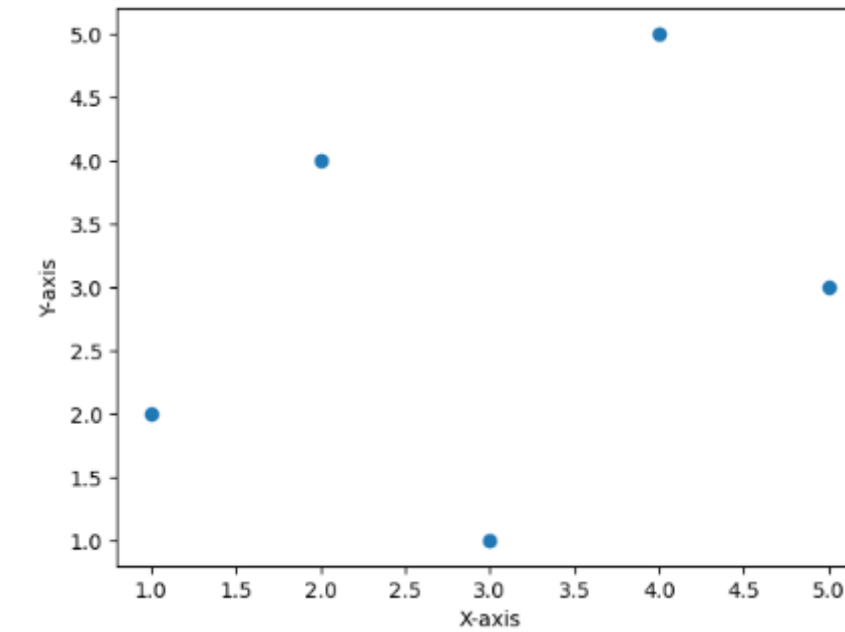
Detecting Outliers



Boxplot



Histogram



Scatterplot

Z-Score (Standard Score)

$$Z = \frac{X - \mu}{\sigma}$$

where: X = Individual data point

μ = Mean of the dataset

σ = Standard deviation of the dataset

IQR (Interquartile Range Method)

$$IQR = Q3 - Q1$$

where: $Q1$ = First quartile (25th percentile)

$Q3$ = Third quartile (75th percentile)

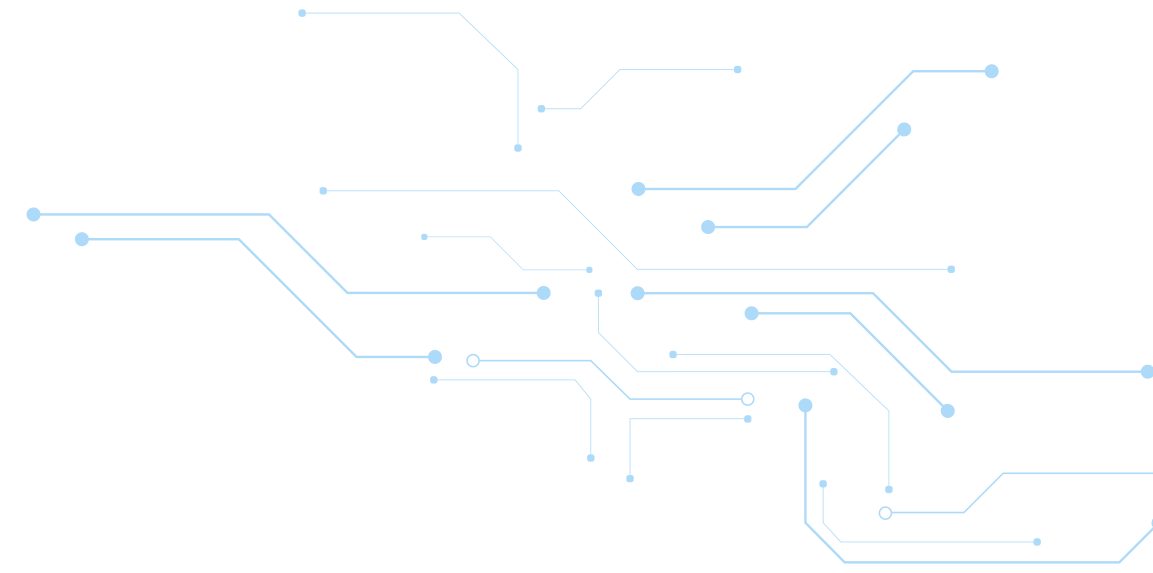
IQR = Interquartile range (spread of the middle 50% of the data)

Handling Outliers

Outliers can distort machine learning models, so handling them effectively is crucial.

Situation	Best Handling Method
Data entry errors	Remove
Right-skewed data	Log Transformation
Outliers are informative	Winsorization or Clipping
Small dataset	Replace with Median

Encoding Categorical values

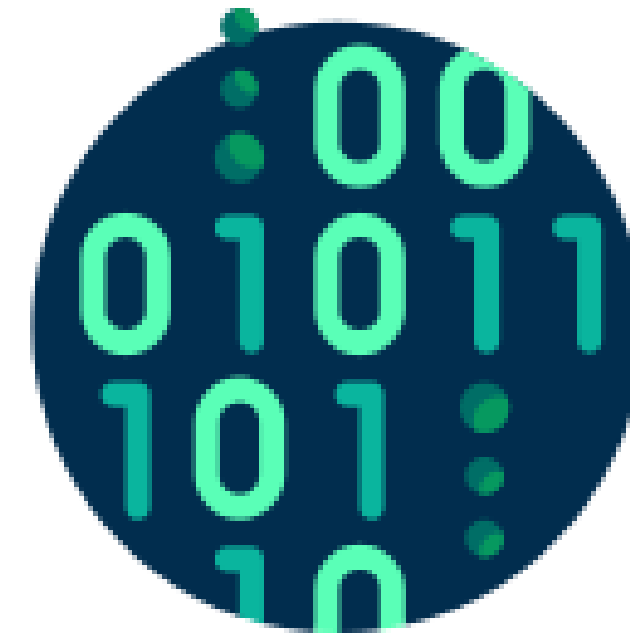


Categorical variables, which represent qualitative data such as names, labels, or groups, must be converted into numerical formats for machine learning algorithms.

One-Hot Encoding

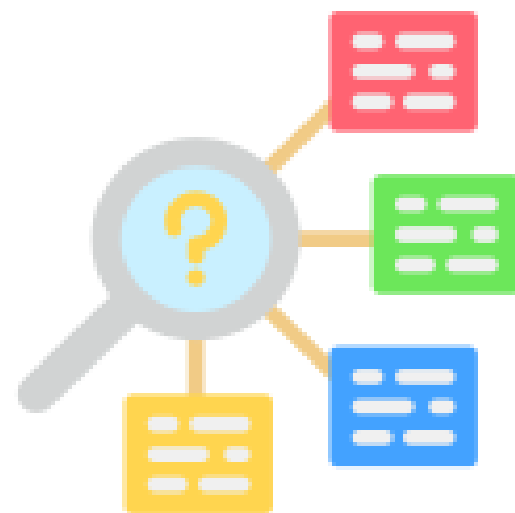


One-Hot Encoding



One-Hot Encoding is a technique used to transform categorical variables into a binary representation. It generates a new binary column for each unique category, assigning a value of 1 to indicate presence of the category and 0 for its absence.

Label Encoding

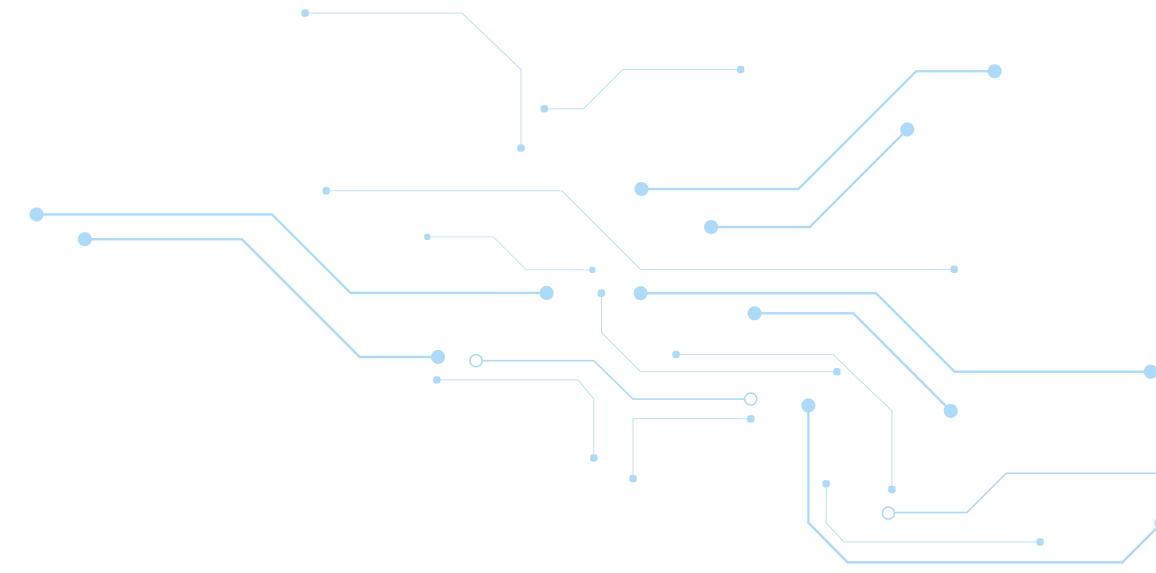


Label Encoding



Label Encoding is a method used to transform categorical data into numerical values, enabling machine learning models that require numerical inputs to process the data. It serves as a crucial preprocessing step in preparing data for a machine learning project.

Feature Scaling



Aspect	Normalization	Standardization
Definition	Scales data to a fixed range (e.g., [0,1]).	Centers data to mean = 0, variance = 1.
Sensitivity	Affected by outliers.	Less sensitive to outliers.
Use Case	Neural networks, KNN.	PCA, regression models.

Splitting Data: Train-Test and Validation

Splitting a dataset into training and testing sets to evaluate the performance of a machine learning model.

Train Dataset: Used to train or "fit" the machine learning model.



Test Dataset: Used to evaluate the performance of the trained machine learning model on unseen data.

Handling Missing Values and Outliers (Demonstration)

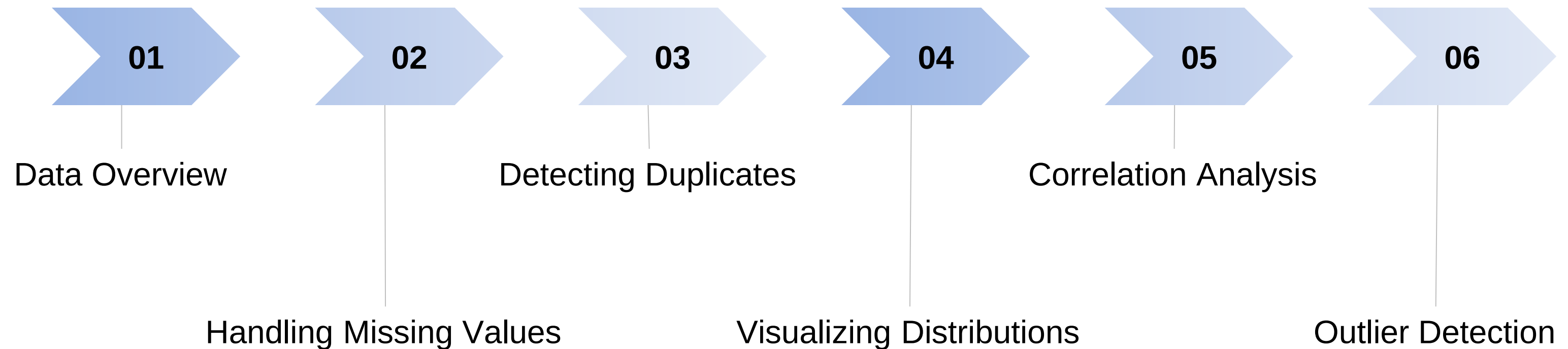
Note: Refer to the Module 1: Demo 1 on LMS for detailed steps.

Exploratory Data Analysis

What is EDA?

Exploratory Data Analysis (EDA) is the process of analyzing and visualizing a dataset to understand its structure, detect patterns, identify anomalies, and summarize key characteristics before applying machine learning models.

EDA Process:



Why EDA is Necessary?

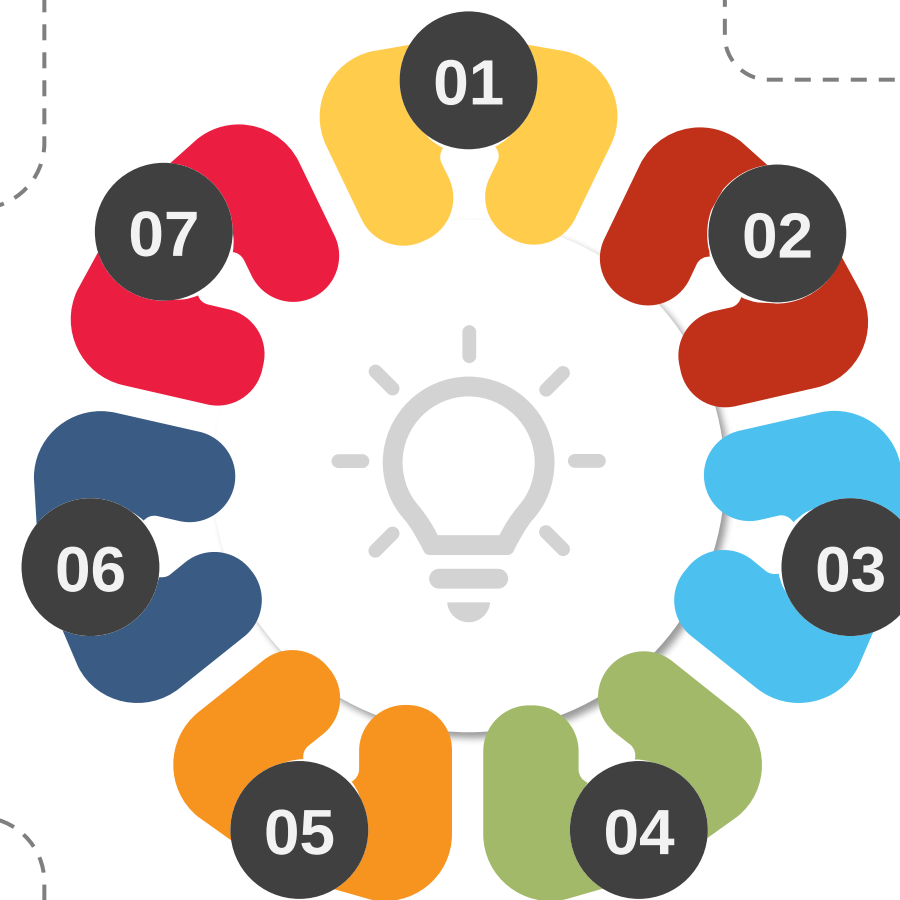


Selecting the Right Model & Features:

Helps determine feature importance and suitable algorithms.

Improving Model Performance: Ensures cleaner data, leading to better training and accurate predictions.

Feature Relationships: Reveals correlations and dependencies between variables.



Understanding Data Structure: Helps identify data types, distributions, and patterns.

Detecting Missing Values: Ensures proper handling of incomplete data to prevent biases.

Identifying Outliers: Helps remove or transform anomalies that could distort model performance.

Spotting Data Imbalance: Ensures fair model training by detecting skewed class distributions.

Mean Imputation

- e! Restores the dataset for further analysis.
- e! Provides a balanced estimate for the missing value.

Time	Launch Pad (°C)
Morning	28
Noon	35
Evening	NaN

Time	Launch Pad (°C)
Morning	28
Noon	35
Evening	31.5

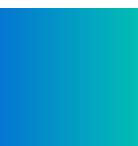


Median Imputation

- e! Reduces the impact of outliers.
- e! Ensures realistic data for trend analysis.

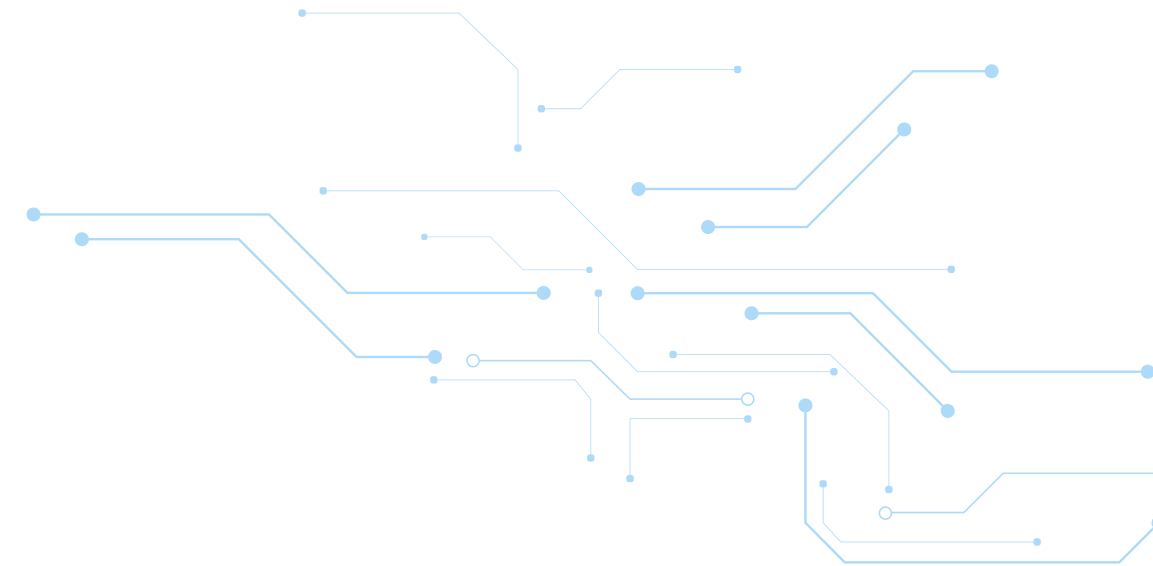
Time	Mission Control (°C)
Morning	NaN
Noon	37
Evening	34

Time	Mission Control (°C)
Morning	35.5
Noon	37
Evening	34



Mode Imputation

- e! Preserves the most common trend in the data.
- e! Ensures consistency in recurring conditions.



Time	Rocket Assembly Zone (°C)
Morning	24
Noon	NaN
Evening	24

Time	Rocket Assembly Zone (°C)
Morning	24
Noon	24
Evening	24

Univariate Analysis

- e! Univariate imputation fills missing data based on a single variable, using methods like mean, median, or mode
- e! This method treats each missing value independently.

Time	Launch Pad Temperature (°C)
Morning	30
Noon	35
Evening	NaN

Time	Launch Pad Temperature (°C)
Morning	30
Noon	35
Evening	32.5

Bivariate Analysis

- e! Bivariate imputation estimates the missing value based on the relationship between two variables.
- e! It leverages correlation or other statistical techniques to predict missing values.

Time	Temperature (°C)	Humidity (%)
Morning	NaN	60
Noon	35	55
Evening	33	65

Time	Temperature (°C)	Humidity (%)
Morning	34	60
Noon	35	55
Evening	33	65

Multivariate Analysis

- e! Multivariate imputation fills in missing values based on multiple variables in the dataset.
- e! This technique considers the interdependence of several features, often using regression models or machine learning algorithms to predict missing data.

Time	Temperature (°C)	Pressure (Pa)	Humidity (%)
Morning	28	NaN	60
Noon	35	1015	55
Evening	30	1012	65

Time	Temperature (°C)	Pressure (Pa)	Humidity (%)
Morning	28	1013	60
Noon	35	1015	55
Evening	30	1012	65

Detecting Data Imbalance

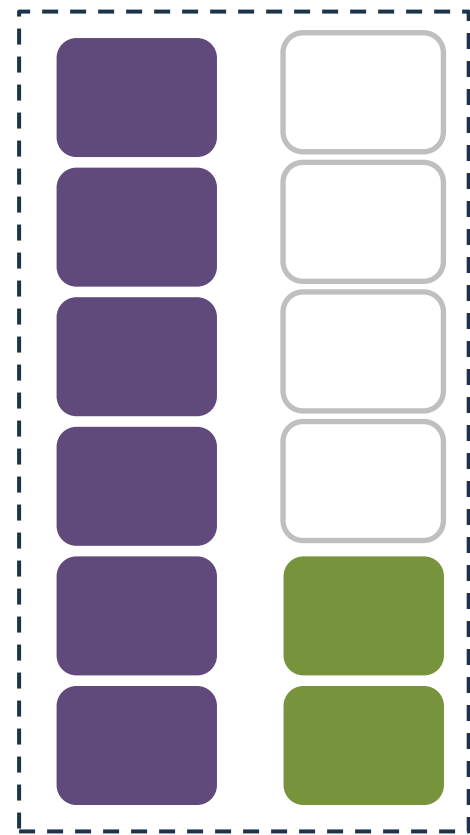
Class imbalance can negatively impact machine learning models by causing biased predictions. Here are the key methods to detect it in ML.

Methods:

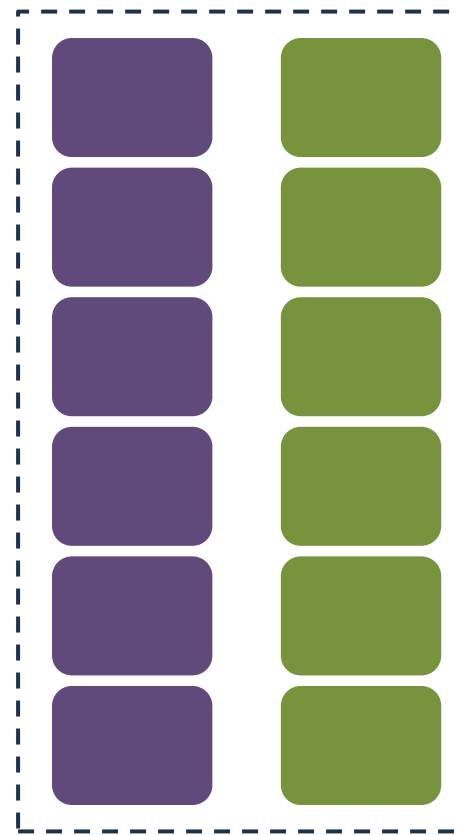
- e! Class Distribution Check
- e! Bar Chart Visualization
- e! Imbalance Ratio (IR)
- e! Confusion Matrix
- e! Precision, Recall, and F1-Score
- e! ROC-AUC Score

ID	Feature 1	Feature 2	Class (Target)
1	2.5	1.3	0 (Majority)
2	3.1	2.2	0 (Majority)
3	1.8	0.9	0 (Majority)
4	2.7	1.5	0 (Majority)
5	3.0	2.1	1 (Minority)

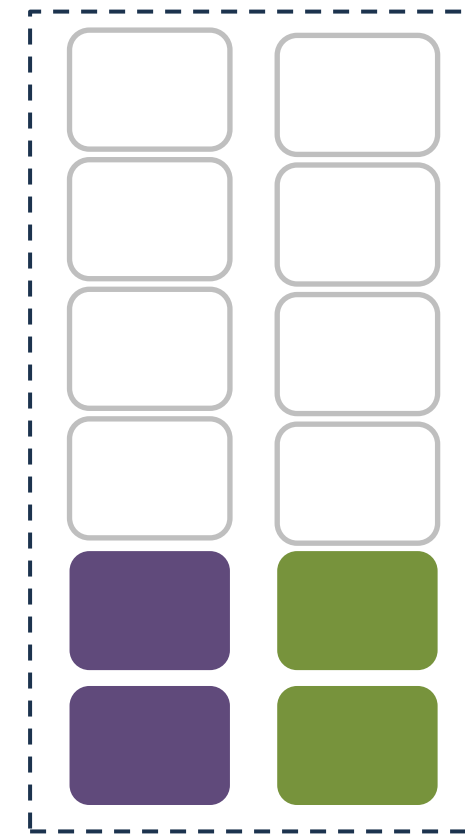
Oversampling and Undersampling



Original Dataset



Over-sampled Data



Under-sampled Data

Oversampling increases the number of minority class samples to balance the dataset, often using techniques like SMOTE to generate synthetic data.

Undersampling reduces the number of majority class samples to balance the dataset, removing redundant data but risking information loss.

Scaling Features Using StandardScaler and MinMaxScaler (Demonstration)

Note: Refer to the Module 1: Demo 2 on LMS for detailed steps.

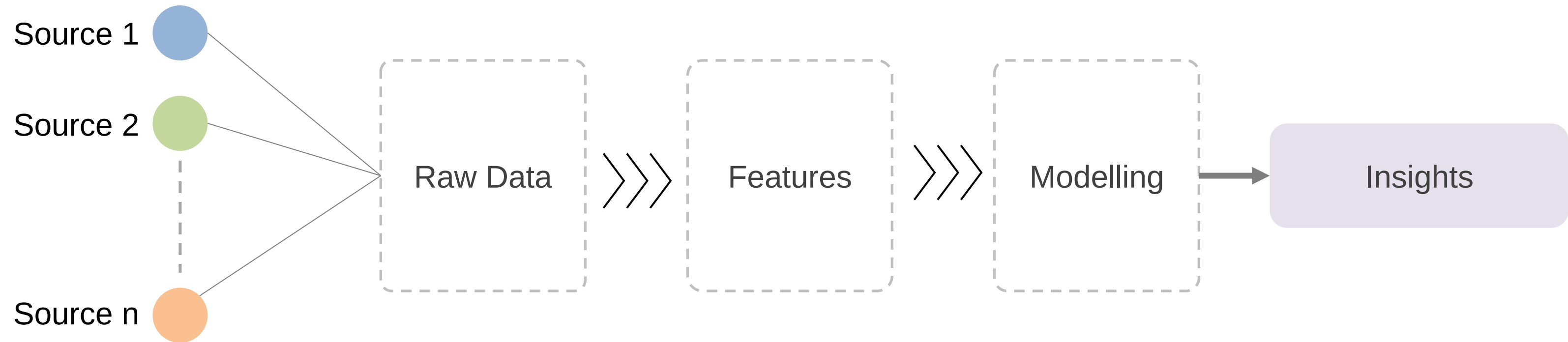
Feature Engineering and Transformation

What is Feature Engineering?

Feature engineering is the process of refining raw data by creating or altering features to optimize a machine learning model's performance.



Creating New Features from Raw Data

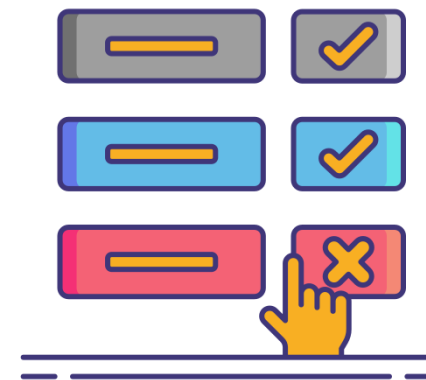


In banking fraud detection, a derived feature like "number of high-value transactions in the past week" can be a strong fraud indicator.

Feature Selection Overview

Feature selection is the process of choosing the most relevant variables for model training while removing redundant or irrelevant features. It helps in reducing overfitting, improving interpretability, and enhancing computational efficiency.

Makes data more interpretable



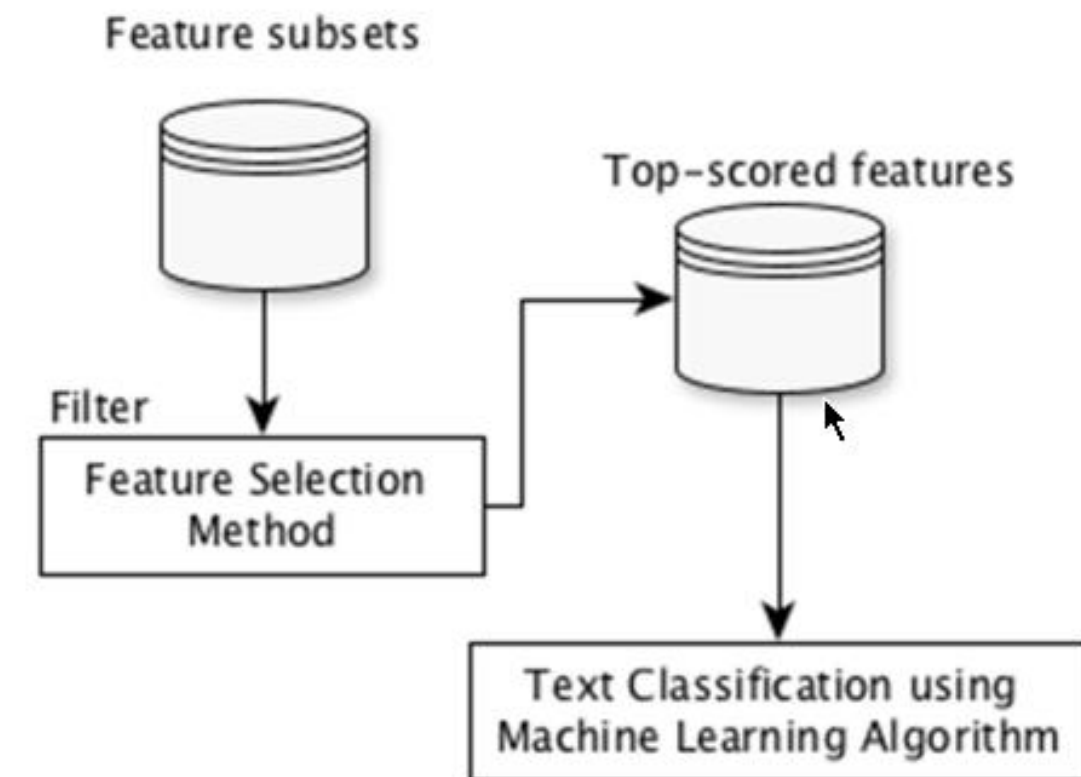
Handles skewed distributions

Enhances relationships between variables

Filter Methods

These methods evaluate feature importance based on statistical measures and select features independently of any model.

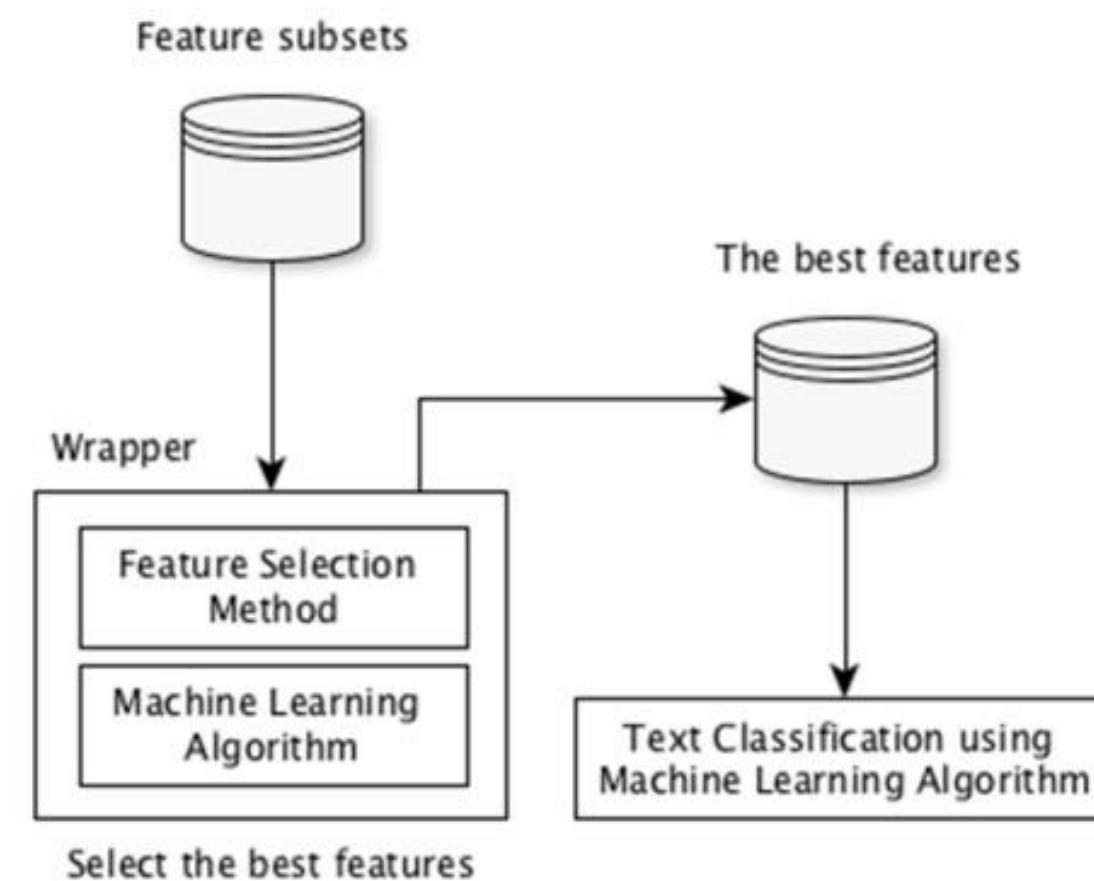
Correlation coefficient
Chi-Square test
Variance Threshold



Wrapper Methods

Wrapper methods assess feature subsets by training a model and evaluating performance iteratively.

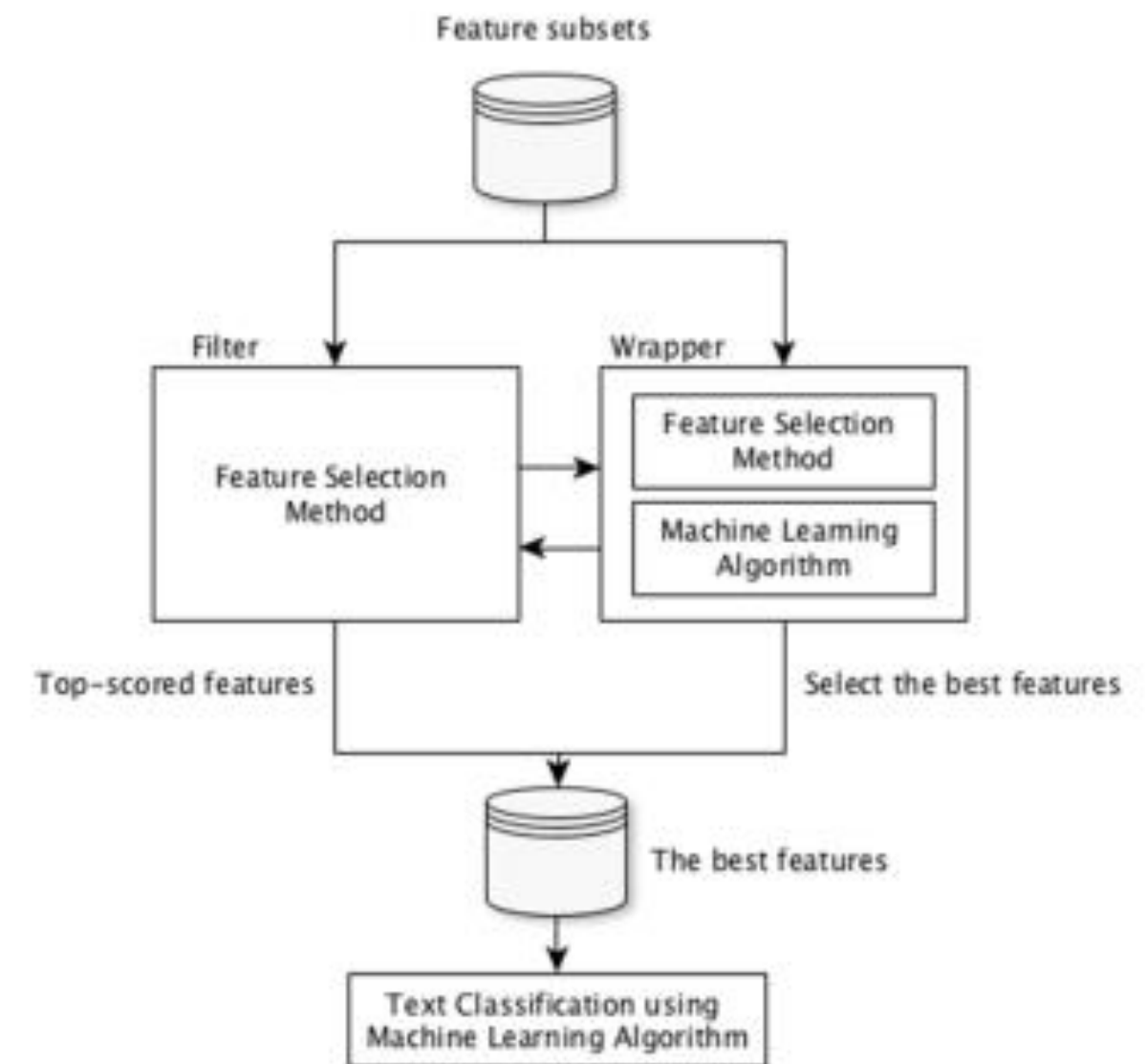
Forward Selection
Backward Elimination
Recursive Feature Elimination



Embedded Methods

These methods perform feature selection during model training by assigning importance scores to features.

LASSO (L1 Regularization)
Decision Tree-based feature importance
Elastic Net



Handling Multicollinearity (VIF)

- e! Variance Inflation Factor is the ratio of the variance of a feature's coefficient divided by the feature's coefficient variance when fit by itself.
- e! Detects multicollinearity when there is a correlation between the predictors.
- e! It is calculated by the R^2 value, which is substituted into the VIF formula,

Interpreting the VIF :

- e! If $VIF = 1$ then, the predictors are not correlated.
- e! If $1 < VIF < 5$ then, the predictors are moderately correlated.
- e! If $VIF > 5$ then, the predictors are highly correlated.

$$VIF = \frac{1}{1 - R_i^2} \text{ where } i = x_1, x_2, x_3, \dots, x_n$$

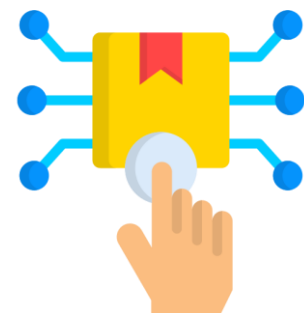
Usually, VIF from 2 to 2.3 is preferable to check the correlation between predictors.

Introduction to Feature Transformation

Feature transformation modifies existing features to enhance their representation for a machine learning model.



Raw Data



Feature Transformation



Improved Data



Better Model Performance

Log and Binning Transformations

Log

Handles skewed distributions by converting data into a more normal shape.

Person	Income (Before Log)	Income (After Log)
A	\$30,000	10.3
B	\$50,000	10.8
C	\$200,000	12.2
D	\$500,000	13.1

Bin

Converts continuous variables into categories to simplify patterns

Customer	Age (Continuous)	Age Group (Binned)
1	23	Young Adult
2	35	Adult
3	60	Senior
4	45	Adult

Interaction Features

Creates new features by combining two or more existing ones to capture hidden relationships.

You are analyzing a digital marketing campaign and want to measure how often users click on an ad after seeing it.

$$CTR = \frac{\text{Clicks}}{\text{Impressions}} \times 100$$

Where:

Clicks = Number of times users clicked the ad

Impressions = Number of times the ad was shown

Ad Campaign	Clicks	Impressions	CTR (%)
A	50	1000	5%
B	200	5000	4%
C	75	1500	5%

Dimensionality Reduction: PCA

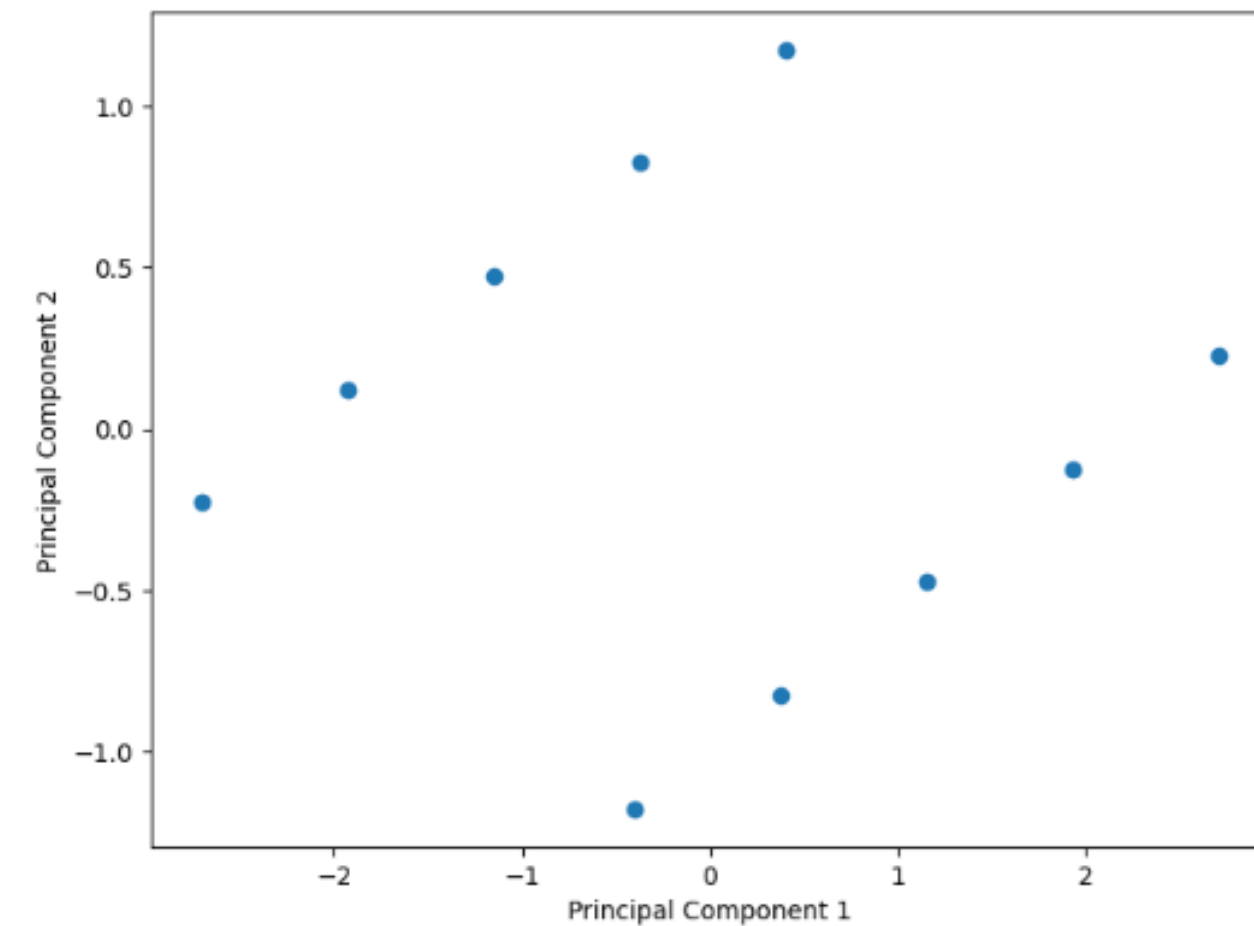
PCA (Principal Component Analysis) is a linear dimensionality reduction technique that transforms correlated features into a set of uncorrelated principal components, ordered by variance.

1. Linearity

2. Computation

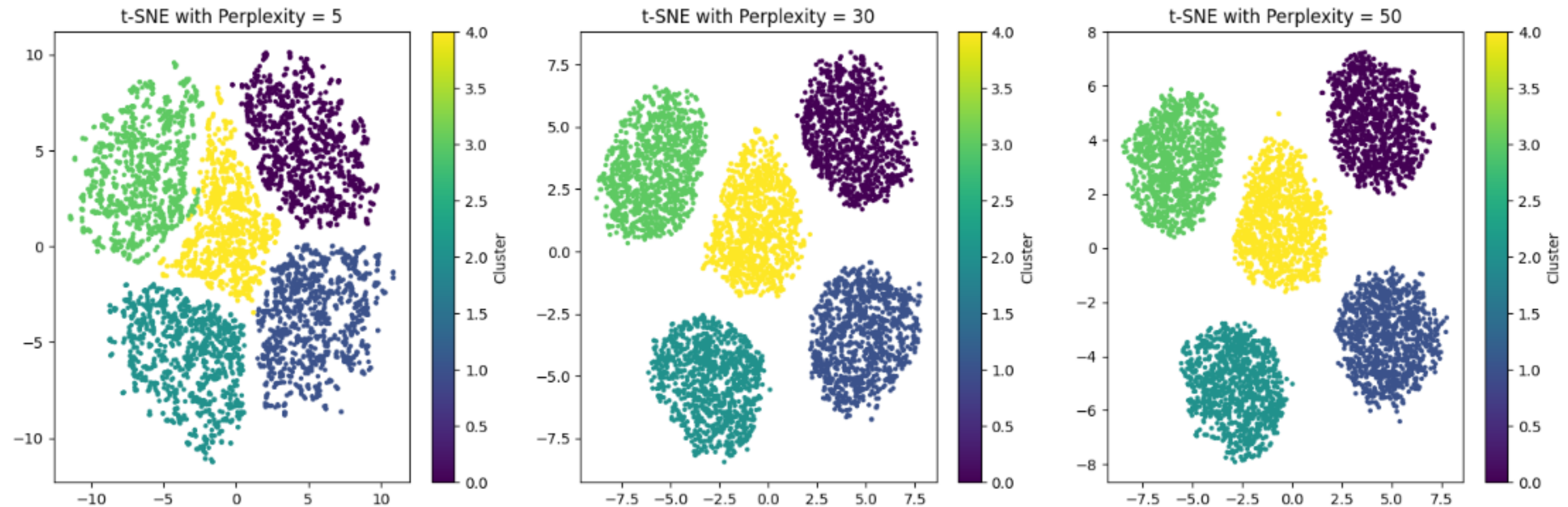
3. Information Preservation

4. Feature Extraction



Dimensionality Reduction: t-SNE

t-SNE is a non-linear dimensionality reduction technique primarily used to visualize high-dimensional data.



Handling Multicollinearity

Variance Inflation Factor (VIF)

- e! VIF is the ratio of the variance of a feature's coefficient divided by the feature's coefficient variance when fit by itself.
- e! Detects multicollinearity when there is a correlation between the predictors.
- e! It is calculated by the R^2 value, which is substituted into the VIF formula,

$$\text{VIF} = \frac{1}{1 - R_i^2} \text{ where } i = x_1, x_2, x_3, \dots, x_n$$

Interpreting the VIF :

- e! If $\text{VIF} = 1$ then, the predictors are not correlated.
- e! If $1 < \text{VIF} < 5$ then, the predictors are moderately correlated.
- e! If $\text{VIF} > 5$ then, the predictors are highly correlated.

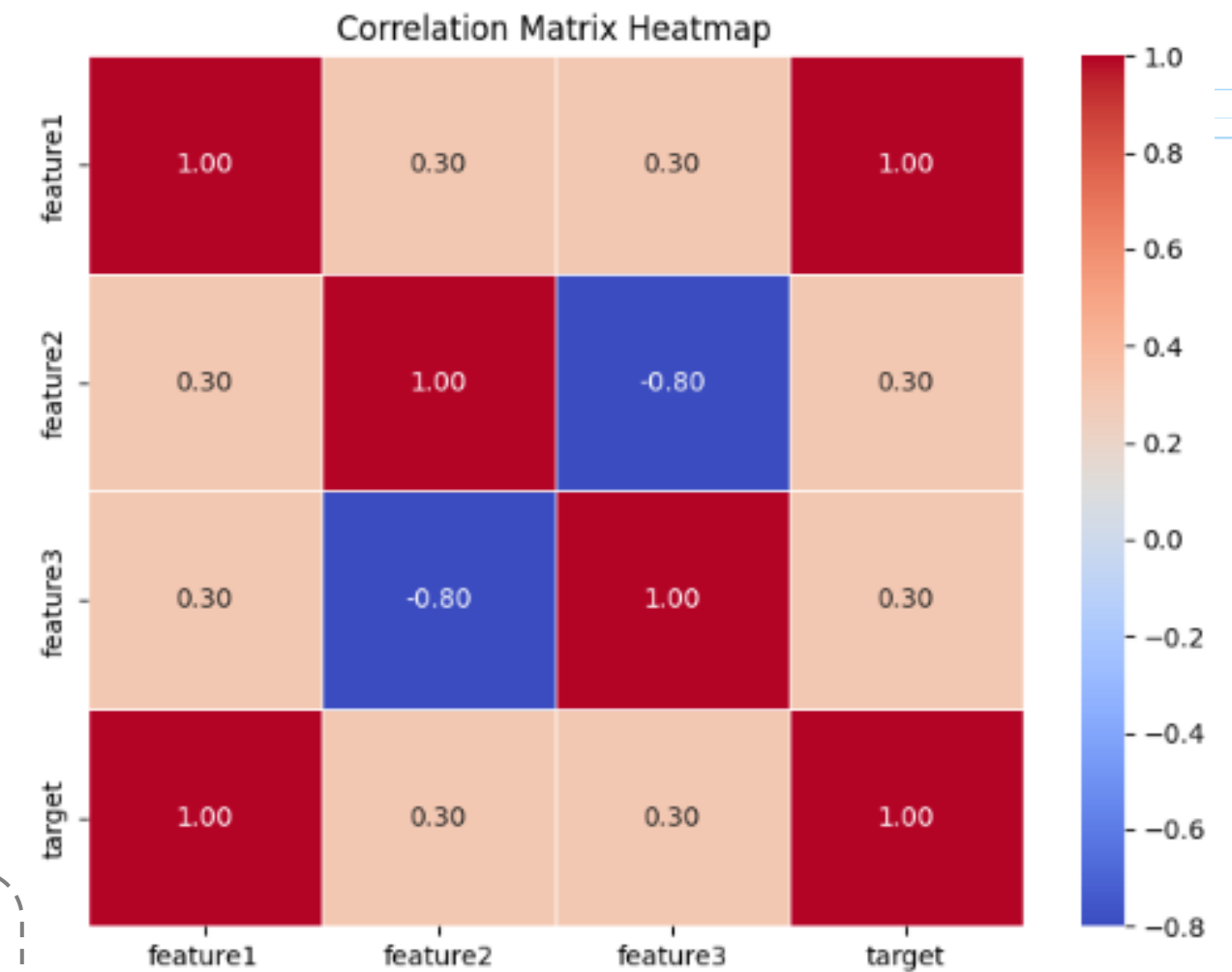
Usually, VIF from 2 to 2.3 is preferable to check the correlation between predictors.

Correlation Matrix

A correlation matrix shows the relationships between different variables in a dataset.

Values range from -1 to 1:

- +1 → Strong positive correlation (as one increases, the other increases).
- 1 → Strong negative correlation (as one increases, the other decreases).
- 0 → No correlation.



Encoding Categorical Variables (Demonstration)

Note: Refer to the Module 1: Demo 3 on LMS for detailed steps.

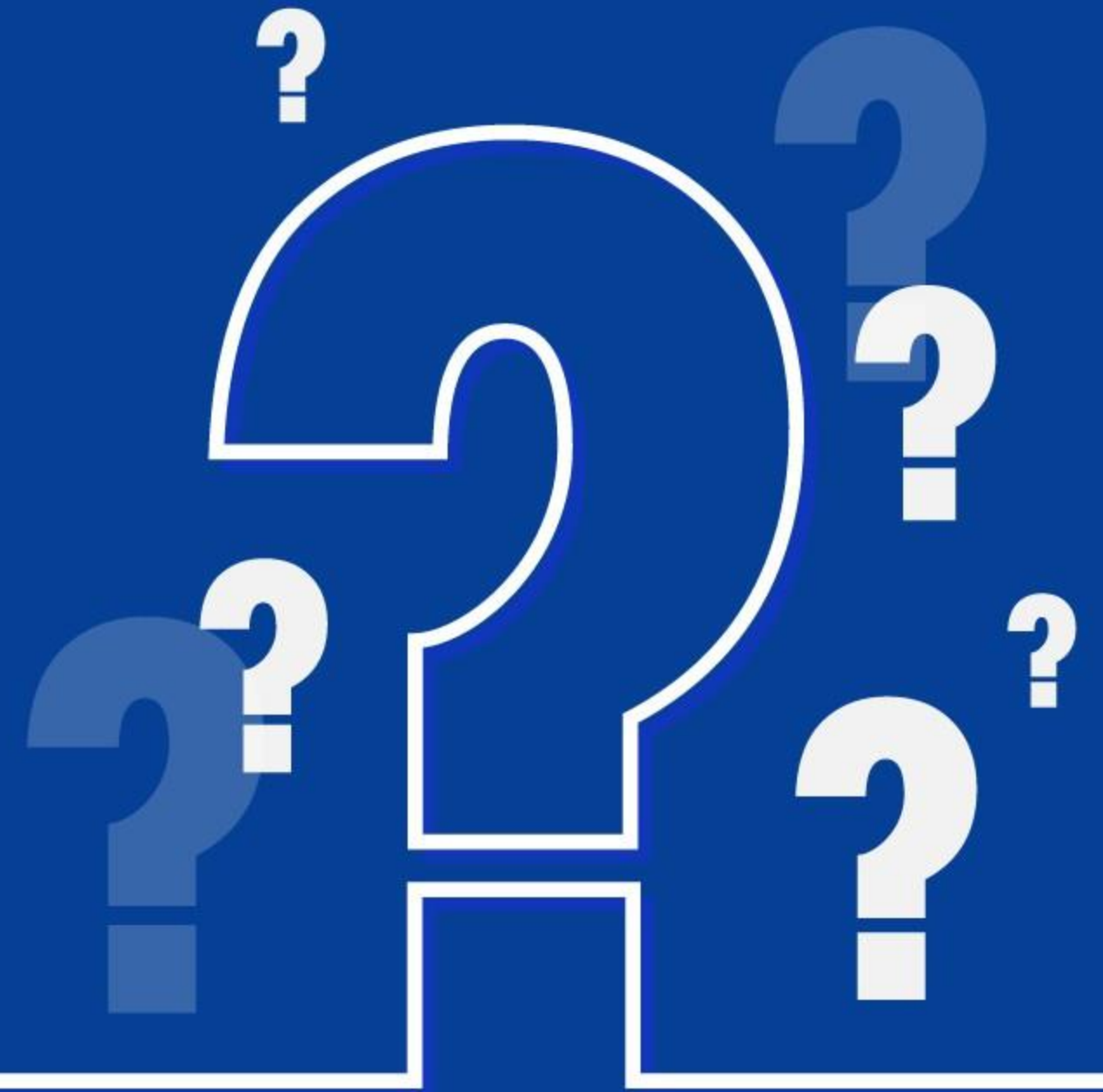
Summary

In this lesson, you have learned to:

- e! Describe Machine Learning, its types, and fundamental concepts.
- e! Explain the significance, scope, and key applications of Machine Learning.
- e! Implement data preprocessing techniques, including handling missing values, encoding categorical variables, and feature scaling.
- e! Analyze datasets using Exploratory Data Analysis (EDA) to identify patterns, outliers, and imbalances.
- e! Apply feature engineering methods, such as selection, transformation, and addressing multicollinearity.



Questions





Feedback





Thank You

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